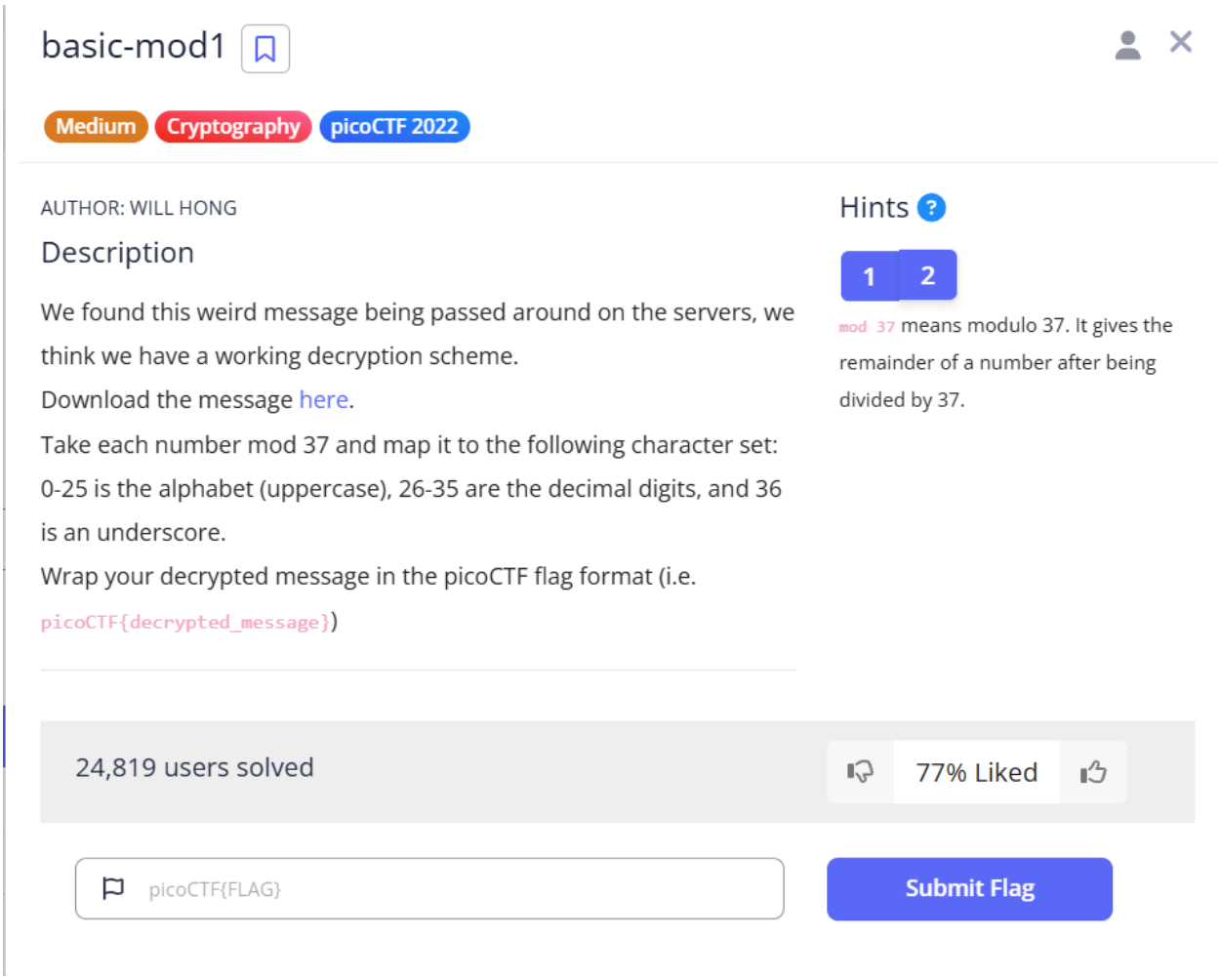
Saint Columban College
College of Computing Studies
Pagadian City

Basic-mod1



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Challenge: Basic-mod1

Category: Cryptography

Date: 03/22/26

I. Objective

The objective of this challenge is to understand how **modular arithmetic (mod)** can be used to encode and decode messages.

The goal is to take encoded numerical values, apply **modulo 37**, and convert the results into readable characters to reveal the hidden flag.

II. Background

In cryptography, numbers are often used to represent letters and symbols. One common technique is **modular arithmetic**, where numbers are reduced within a fixed range using a modulus.

In this challenge:

- The modulus used is **37**
- This means all numbers are mapped within the range **0–36**

Each number corresponds to a character such as:

- Letters (a–z)
- Numbers (0–9)
- Special characters (like _)

To decode the message:

1. Take each number
2. Apply **mod 37**
3. Convert the result into its corresponding character

This is a basic introduction to how encoding and decoding works in **cryptography**.

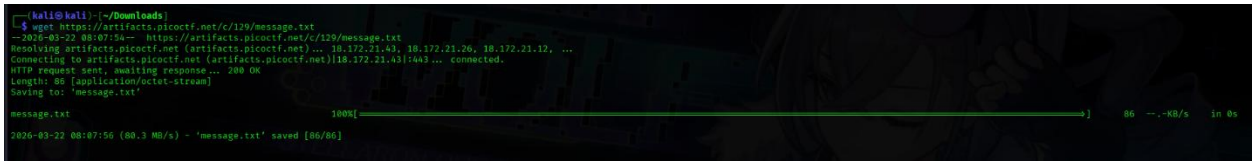
III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Used as the working environment
- **Sublime Text (subl)** - Used to view and edit files
- **Python 3** - Used to automate decoding of the message

IV. Methodology

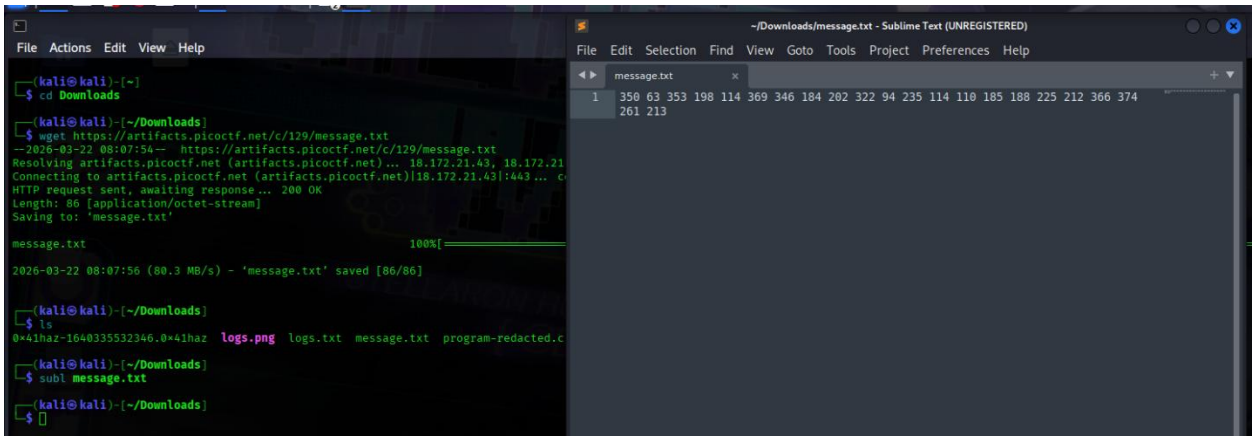
1. Launched **Kali Linux**.
2. Downloaded the provided file using:

wget <link_address>



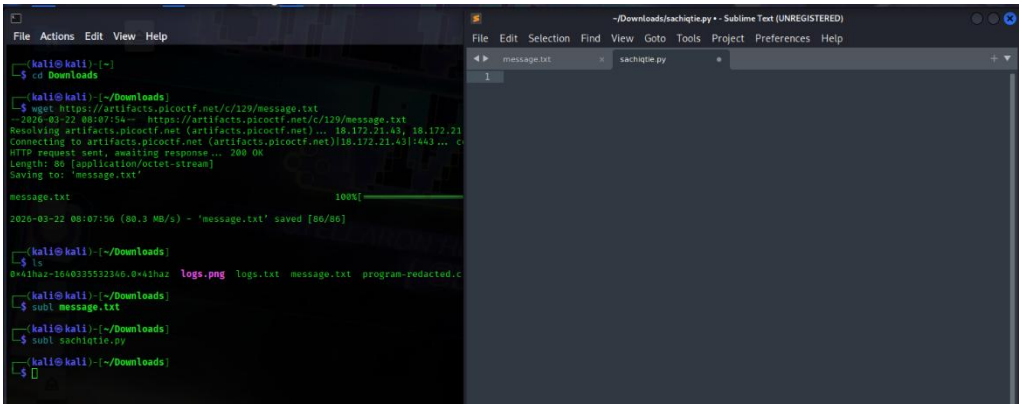
3. Opened the encoded message using:

subl message.txt



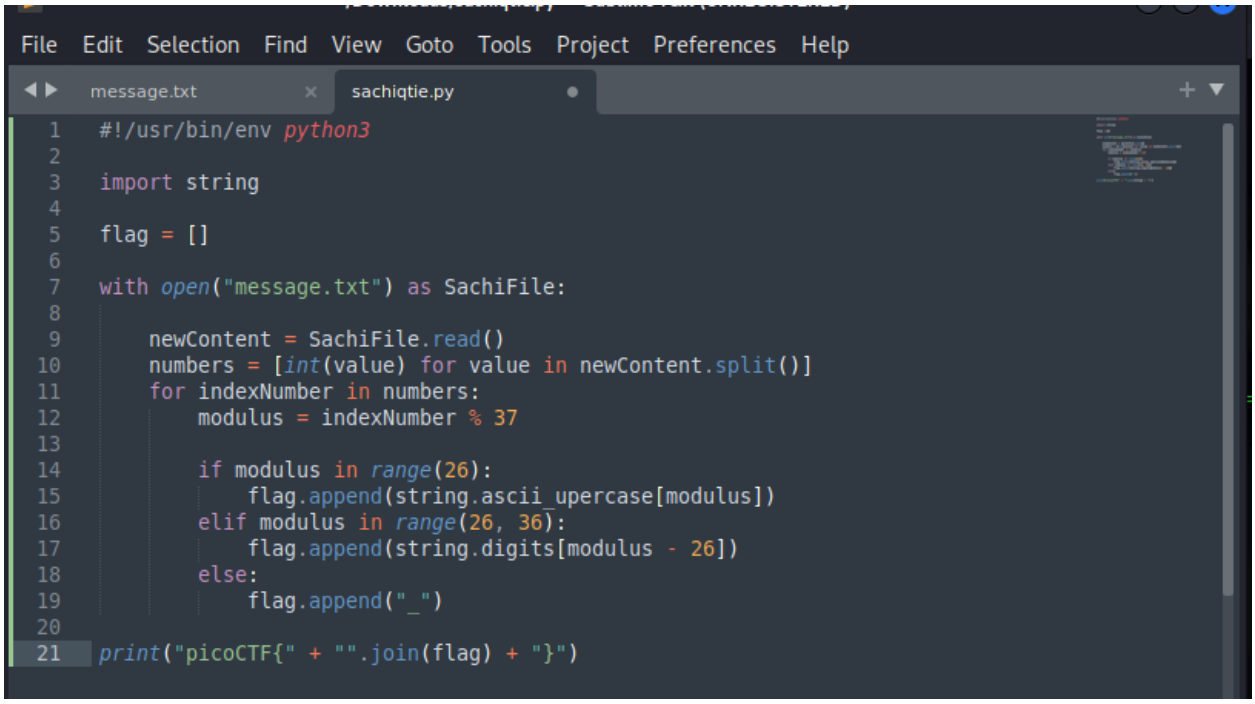
4. Created a Python script to decode the message:

subl <filename>.py



I like naming my file with the things I like/love.

5. Wrote a Python program that:
   Reads the encoded numbers
   Applies **modulo 37**
   Converts each result into its corresponding character



6. Ran the program (in Sublime: **Ctrl + B**) or using:
   python3 <filename>.py

I used CTRL + B in my case.

7. The decoded output revealed the **flag**.

```
picoCTF{R0UND_N_R0UND_ADD17EC2}
[Finished in 30ms]
```

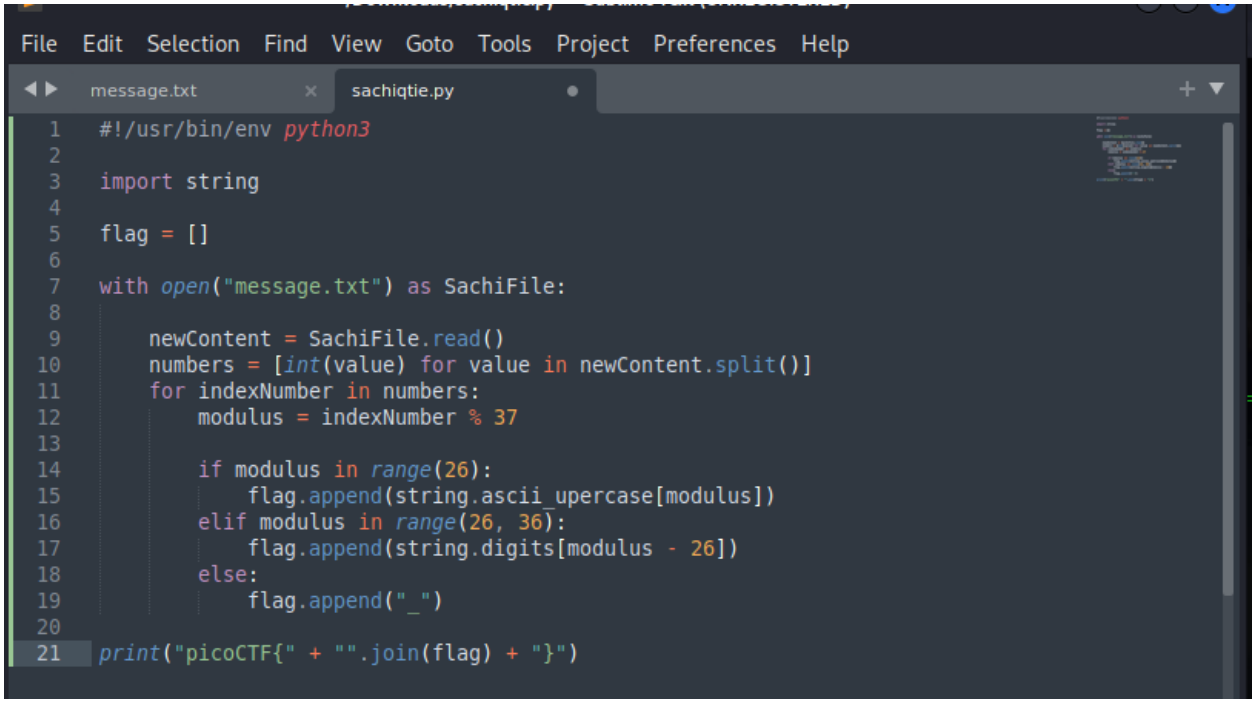
V. Result

picoCTF{R0UND_N_R0UND_ADD17EC2}

```
picoCTF{R0UND_N_R0UND_ADD17EC2}
[Finished in 30ms]
```

VI. Explanation

Let's analyze this script:



Let's try to explain it line by line…

1. #!/usr/bin/env python3

This line is called a **shebang**

So… What does **#!** mean?

- **"#"** is normally a comment in Python
- **BUT** when it's at the very first line with **"!",** it becomes special

It tells the operating system (Linux):
"Hey, use this program to run this file"

So... breaking it down:
#!

Means:
"This file should be executed using the following program"

2./usr/bin/env

This is a **program in Linux**.

Its job:
"Find where a command/program is installed"

Think of it like:
A **search tool** that looks for programs in your system

3.python3

This is the program we want to use.
So the full line means:
"***Use env to find python3, then run this script using it"***

Why not just write this?

#!/usr/bin/python3

That would also work, BUT:

- Not all systems have Python in /usr/bin/python3
- Some systems install Python somewhere else

So using:

#!/usr/bin/env python3

makes your script **portable** (works on more systems)

Why is this important?

Without shebang, you must run:

Python3 decode.py

With shebang + executable permission:

Chmod +x decode.py

./decode.py

4. **import string**

Imports a built-in Python module.

Gives access to:

- string.ascii_uppercase → ABCDEFGHIJKLMNOPQRSTUVWXYZ
- string.digits → 0123456789

5. **flag = []**

Creates an **empty list**.

This will store decoded characters one by one.

6. **with open("message.txt") as SachiFile:**

Opens the file safely.

- "message.txt" → the file
- SachiFile → variable name

with ensures:

- File is **automatically closed** after use

7. **newContent = SachiFile.read()**

Reads the whole file as text.

8.**numbers = [int(value) for value in newContent.split()]**

This is a **list comprehension** (shortcut loop).

Step-by-step:

- .split() → splits text by spaces
- int(value) → converts each into a number

result:

[1, 2, 3, 4, 5]

9. **for indexNumber in numbers:**

Loops through each number.

Like:

- first → 54
- next → 12

10. **modulus = indexNumber % 37**

Applies **modulo 37**

Keeps values between:

0 → 36

Example:

- 54 % 37 = 17
- 99 % 37 = 25

11. **if modulus in range(26):**

Checks if value is:

0 → 25

These represent **letters A–Z**

12. flag.append(string.ascii_uppercase[modulus])

Adds a letter to the list.

Example:

- **0 → A**
- **1 → B**

13. elif modulus in range(26, 36):

Checks if value is:

26 → 35

These represent digits (0–9)

14. flag.append(string.digits[modulus - 26])

Why - 26?

Because:

- **26 should map to digit 0 (based in description)**
- **27 → 1**

Fixes the index.

15. else:

Handles the last value:

36

16. flag.append("_")

If value = 36 → underscore _

17. print("picoCTF{" + "".join(flag) + "}")

Final output:

- "".join(flag) → combines letters into a word
- Adds picoCTF{} format

That should wrap things up in the script… Is it understandable?

So…

What is "mod" (modulo)?

"Mod" means **remainder after division**.

Example:

- 10 mod 3 = 1 (because 10 ÷ 3 leaves remainder 1)

Why mod 37?

Idk? Maybe because there are **37 possible characters**:

- 26 letters (A–Z)
- 10 digits (0–9)
- 1 special character (_)

So every number gets mapped into this range using mod 37 ig.

How does decoding work?

Think of it like a **code wheel**:

- Each number points to a character
- If the number is too big, we "wrap it around" using mod

Example:

- 40 mod 37 = 3
- So we use character at index **3**

Why use Python?

Instead of decoding manually (which is slow and error-prone), Python:

- Automates the process
- Quickly converts all numbers into characters
- Ensures accuracy

VII. Conclusion

This challenge demonstrates how **modular arithmetic is used in cryptography to encode and decode messages**.

By applying **mod 37** and mapping numbers to characters, the hidden message was successfully decoded. Using Python made the process efficient and reliable.

This reinforces an important concept in cybersecurity:

Mathematics is a fundamental tool in encryption and decryption.

— The Analyst: Hyposelenia